

TANUJ MANGALAM

Computer Engineering | Purdue University | Degree: December 19, 2026

AI / ML · Analog IC Design · Semiconductor Devices · Embedded Systems

(317) 645-8822 · tmangala@purdue.edu · [linkedin.com/in/tanuj-mangalam-474459233](https://www.linkedin.com/in/tanuj-mangalam-474459233) · github.com/tanujmangalam · US Citizen | GPA: 3.26

Engineering Portfolio

Six technical projects spanning AI/ML research, analog IC design, and embedded systems

PROJECT 1

Parkinson's Cognitive Assessment ML

Clinical ML | Wearable Sensors | RMSE 2.17

1

PROJECT 2

Diabetic Retinopathy CNN Classifier

Medical Imaging | 7-Layer CNN | 60% Accuracy

2

PROJECT 3

Real-Time Gaze Estimation System

Computer Vision | Sim2Real | RL | Grade: A

3

PROJECT 4

Differential CMOS Op-Amp with SC-CMFB

Analog IC | 45.6µW | 39.2MHz GBW | GPDK045

4

PROJECT 5

MOSFET Device Characterization

Semiconductor Devices | First-Principles | ECE 305

5

PROJECT 6

Atari Breakout Embedded System

Embedded C | SPI | PWM | Register-Level

6

TECHNICAL SKILLS SUMMARY

Languages: Python · C/C++ · SystemVerilog · MATLAB · SQL · JavaScript
AI/ML: TensorFlow · Keras · PyTorch · XGBoost · CNNs · Ridge/ElasticNet · LLMs
Hardware: CMOS IC Design · SPICE · Cadence · NANOMOS · GPDK045 · SPI · PWM · SystemVerilog
Fabrication: RIE Dry Etching SOP · IC/MEMS · Cleanroom Protocols · Wafer Handling

EDUCATION

B.S. Computer Engineering — AI/ML & Microelectronics/Semiconductors (Dual Concentration)

Purdue University, West Lafayette, IN | GPA: 3.26 | Degree Conferred: December 19, 2026
North Carolina State University (Admitted, Spring 2027)

Clinical ML

Wearable Sensors

Python

Ridge/ElasticNet

PCA

Leave-One-Out CV

XGBoost

OVERVIEW

Designed an end-to-end ML pipeline to predict Montreal Cognitive Assessment (MoCA) scores in Parkinson's patients from wearable TSFEL accelerometer data — enabling remote, continuous cognitive monitoring without clinic visits. Supervised by Prof. Hee Tae Jung and Dr. Sarath Janga at IU Indianapolis.

2.17

Best RMSE

7

Models Tested

Ridge

Top Method

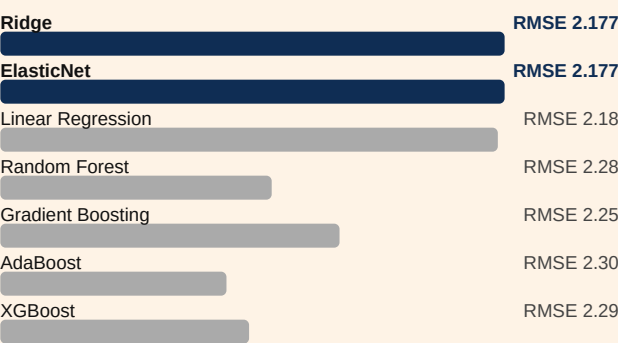
LOO-CV

Validation

METHODOLOGY

- Extracted TSFEL time-series features from 5-minute accelerometer w
- Applied PCA for dimensionality reduction — confirmed variance reduc
- Ridge/ElasticNet regularization to prevent overfitting on high-dimensic
- Leave-One-Out CV — maximizes every patient sample in small clinic
- Diagnosed StandardScaler bug converting MoCA scores to z-scores -
- Collaborated with clinical team to interpret RMSE in clinical context (N

MODEL BENCHMARK RESULTS



KEY ENGINEERING INSIGHT

Ridge/ElasticNet outperformed XGBoost because clinical accelerometer data is high-dimensional with correlated features — L2 regularization's coefficient shrinkage is the correct inductive bias. Recognizing which algorithm fits which data structure is what separates a practitioner from a practitioner who understands why.

OVERVIEW
Built a 3-class retinal image classifier for Diabetic Retinopathy (DR) and Diabetic Macular Edema (DME) detection.
Dataset: MESSIDOR-2 — 1,200 retinal fundus images. Classes: No DR (546) · DR only (428) · DR+DME (226).
Addresses shortage of ophthalmologists through AI-assisted screening at scale.



CNN ARCHITECTURE (KERAS)

Input	32 filters
Conv 1	64 filters + BatchNorm + MaxPool
Conv 2	128 filters + BatchNorm + MaxPool
Conv 3	256 filters + BatchNorm + MaxPool
Conv 4	512 filters + BatchNorm + MaxPool
Conv 5	1,024 filters
Conv 6	2,048 filters
Conv 7	4,096 filters
Dense	Flatten → FC → Dropout → Softmax(3)

MODEL COMPARISON (TEST ACCURACY)

CNN (designed)	60%
Random Forest	57.5%
SVM	55%
KNN	55%
XGBoost	53.75%
Logistic Reg.	52.5%

WHY CNN WON — THE TECHNICAL REASONING
RF and XGBoost operate on flat feature vectors — they lose all spatial information in retinal images. CNNs learn hierarchical spatial features (vessel patterns, lesion shapes, exudates) that define disease stages. Data augmentation (flips, rotations) addressed class imbalance in the DME class (only 226 of 1,200 images). Dropout prevented overfitting on the small medical dataset.

WHAT & WHY

- Real-time line-of-sight estimation via standard webcam — no specialized hardware
- Sim2Real bridges synthetic training (free labeled data) to real webcam deployment
- RL optimizes accuracy across arbitrary head positions and screen location
- Hard real-time constraints: model must run fast enough for live user feedback
- Demonstrates production-grade AI deployment under hardware constraints

KEY ENGINEERING CHALLENGES

- Sim2Real gap: synthetic eyes differ visually from real webcam feeds
- Domain randomization: varied lighting, skin tone, background during training
- RL reward = gaze accuracy; model learns policy for any screen position
- Inference latency constraint: compute budget as hard requirement
- Disentangling head pose from true gaze direction

45.6 μ W

Power

39.2 MHz

GBW

4.986 V/ μ s

Slew Rate

52.6 dB

DC Gain

DESIGN DECISIONS & RATIONALE

- Current mirror topology: maintains high performance at minimal current draw
- Cascoding: multiplies output resistance creating high-impedance nodes —
- SC-CMFB: zero static power CM stabilization — chose over CT-CMFB to improve efficiency
- N=140 PMOS multiplier (W=280 μ m, L=10 μ m): maximizes gm/ID ratio, minimizes area
- L=10 μ m long channel: increases Rout for high DC gain — trades bandwidth for gain
- Benchmarked against TSMC 0.18- μ m CMOS reference designs from IEEE

SIMULATION VERIFICATION RESULTS

Parameter	Achieved	Target	Status
Power Consumption	45.6 μ W	<50 μ W	✓ MET
Unity-Gain BW	39.2 MHz	238.83 kHz	✓ MET
Slew Rate	4.986 V/ μ s	5 V/ μ s	✓ MET
DC Gain	52.6 dB	54.2 dB	Close
Noise Floor	-79 dB	-86 dB	Close

Gain gap: increase cascode gain-boosting. Noise gap: chopper stabilization to shift 1/f noise above corner frequency.

PROJECT 5

MOSFET Characterization & Threshold Voltage Engineering

ECE 36200 — Semiconductor Devices | Spring 2026

Semiconductor Devices

NANOMOS

SPICE

Device Physics

Vth Engineering

First-Principles Validation

0.289 V

Vth Baseline

0.605 V

Vth Engineered

3+ OOM

Ioff Reduction

161.94

SS Analytical

ENGINEERING APPROACH

- Engineered Vth via work function (WF 4.2→4.5) and interface charge (Nint)
- $C_{ox} = 4.72 \times 10^{-7} \text{ F/cm}^2$ from 7nm SiO₂ + 2nm HfO₂ high-k gate stack series
- $C_d = 1.29 \times 10^{-7} \text{ F/cm}^2$ at $N_A = 2 \times 10^{17} \text{ cm}^{-3}$; $C_{it} = 6.73 \times 10^{-7} \text{ F/cm}^2$ from unanneal
- $SS = 60 * (1 + (C_d + C_{it}) / C_{ox}) = 161.94 \text{ mV/dec}$ — matched simulation of 162
- $V_{th} = 0.92 \text{ V}$ derived from Phi_{MS}, depletion charge, trap charge — matche
- DIBL stable 131→130.5 mV/V — confirmed geometry unchanged, only ele

ION / IOFF TRADE-OFF ANALYSIS

- Ioff reduced from $8.76 \times 10^{-10} \text{ A}$ to $3.95 \times 10^{-13} \text{ A}$ — over 3 orders of magnitude
- Ion reduced from $8.41 \times 10^{-6} \text{ A}$ to $5.40 \times 10^{-6} \text{ A}$ — direct consequence of higher Vth
- $I_{on} \propto (V_{gs} - V_{th})^2$ — higher Vth reduces overdrive voltage at fixed Vdd
- SS unchanged (88.5 mV/dec) — capacitive ratio unchanged since geometry preserved
- This is the fundamental Ion/Ioff trade-off — cannot reduce leakage without paying driv
- Demonstrates ability to reason about semiconductor device physics trade-offs quantiti

PROJECT 6

Atari Breakout Embedded Hardware System

ECE 36200 — Microprocessor Systems & Interfaces | Fall 2025

Embedded C

SPI Protocol

PWM

Interrupt-Driven

Register-Level

Hardware/Software Co-Design

Real-Time

TECHNICAL IMPLEMENTATION

- SPI protocol implemented from scratch at register level — no library abstr
- PWM audio output via timer register configuration — frequency and duty c
- Interrupt-driven game loop — precise timing without polling, CPU-efficient
- Register-level peripheral configuration — GPIO, SPI, timer, interrupt contr
- Each subsystem (display, audio, input, game logic) unit-tested independen

HARDWARE/SOFTWARE CO-DESIGN

- Software timing requirements directly drove hardware peripheral configuration choice
- Display refresh rate constraint set SPI clock frequency — hardware and software co-
- Interrupt latency measured and verified against real-time game responsiveness requi
- Full system integration testing after independent subsystem verification

PROFESSIONAL EXPERIENCE

Generative AI Extern — Cognizant (2025)

Developed GenAI and LLM agentic solutions in Python on enterprise production platform. Completed live production project applying LLM techniques to real-world enterprise challenges. Prompt engineering, model integration, agentic workflow design.

Health Informatics Research Intern — Luddy IU Indianapolis (May–Jul 2024)

Built end-to-end Parkinson's cognitive prediction ML pipeline. See Project 1 for full details. Supervised by Prof. Hee Tae Jung and Dr. Sarath Janga.

Research Intern — Medical Image Classification — Luddy IUPUI (May–Aug 2023)

Designed CNN/SVM models for Diabetic Retinopathy and Macular Edema classification. See Project 2 for full details.